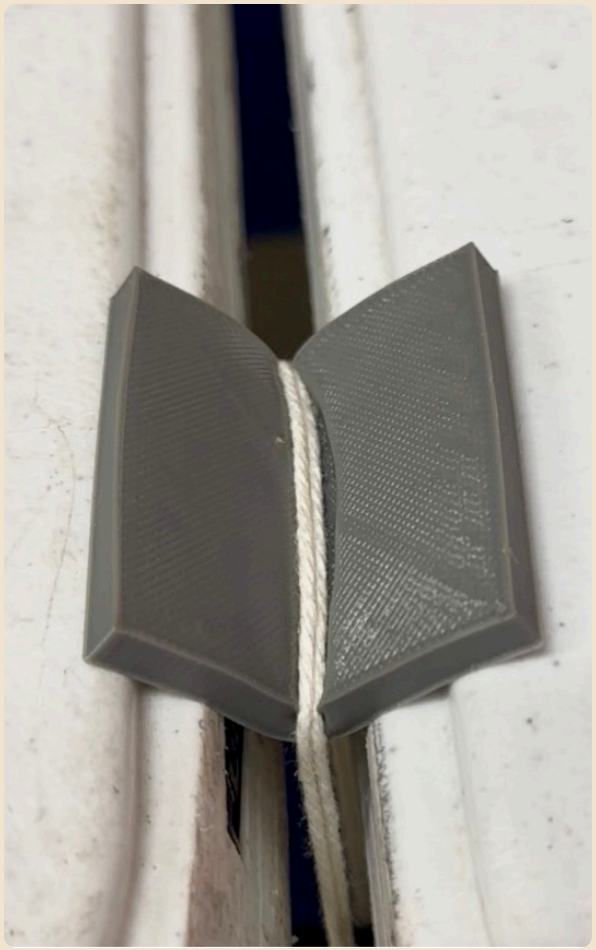
VALUE STATEMENT

PRINTED FOR ONE ASTRONAUT, WHEN THEY NEED IT

OrthoPrint is a custom splint for a lateral ankle ligament tear that a lunar base prints on demand from a scan of the astronaut's leg. Instead of launching a full set of fixed-size splints at about $1,000 per gram, the habitat prints one that fits.



Our printed prototype plates during the hanging-weight test

See the full CAD online
gertrudish978-pixel.github.io/lunar-splint

YOUR DESIGN · OTHER DESIGN FEATURES

SPECS & MATERIALS

ORTHOPRINT	PROTOTYPE #1 · OCT 2026
INJURY	Lateral ankle ligament tear
FIT	Custom, from a limb scan
CLOSURE	Reusable zip-tie style straps, X pattern
SHELL	Solid wall; small vents being re-tested
PRINTER	Habitat FDM or SLA
SIZE	460.59 mm tall; plates 144 mm apart
MASS	To be measured from the final CAD
PRINT TIME	To be measured from the final CAD

PLA
Prototype 1

ABS
Prototype 2

TPU
Prototype 3 · flexes, firm when solid

PETG
Tested · broke when bent

CF-PEEK
The material we hope to use for flight



Printed sample showing the shell thickness

NEXT STEPS

→ Test strap thickness and the load at which straps break
→ Find the temperature where the filament softens
→ Prototype #2 for the Critical Design Review

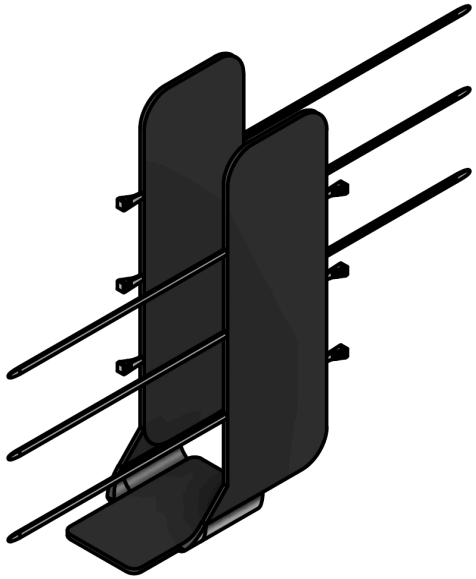
A NASA HUNCH 2026–27 student design project. Not a medical device.

GLADYS PORTER EARLY COLLEGE HIGH SCHOOL
BROWNSVILLE, TX · NASA HUNCH 2026–27

NASA HUNCH · LUNAR SPLINT DESIGN CHALLENGE

ORTHOPRINT

PDR Preliminary Design Review · Oct 21–22, 2026



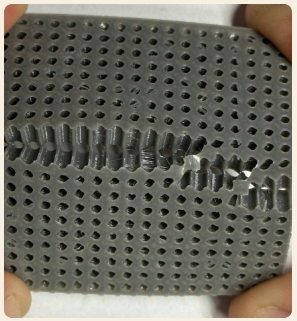
NASA PROJECT	Lunar Splint Design Challenge
PROGRAM	Design & Prototype
SCHOOL	Gladys Porter Early College High School
LOCATION	Brownsville, TX
TEACHERS	Lupita C. Vasquez and Claudia Cortez
TEAM LUNOVA	Gertrudis, Jose and Nancy

Website, videos and CAD
Scan for our testing and prototype videos, CAD and engineering notebook
gertrudish978-pixel.github.io/lunar-splint

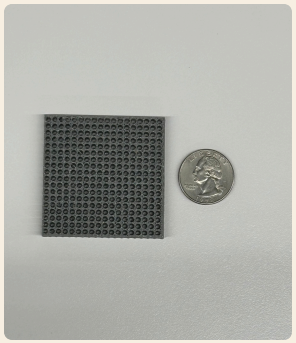
PROTOTYPE HIGHLIGHTS

HOW IT WORKS

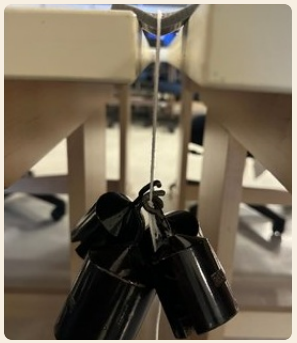
1. **Scan**
   Scan the injured astronaut's lower leg.
2. **Print**
   Print the custom shell and straps on the habitat printer.
3. **Fit**
   Unlatch the splint and fit it on vertically.
4. **Secure**
   Wrap the straps in an X, three times around the ankle.



Sample with holes split, so the wall went solid



TPU sample, quarter for scale



Hanging weights on a sample



Solid TPU, the wall we chose

TEST RESULTS

PETG, holes: broke after bending	REJECTED
TPU, holes: layers split	REJECTED
Solid TPU: firm, moulds slightly	CHOSEN
Straps at key points: ankle stayed still	CHOSEN

YOUR DESIGN · UNIQUE ASPECTS

BUILT FOR THE MOON

INNOVATION 1

SINGLE-PIECE RATCHET CLOSURE

Zip-tie style straps with no loose parts, tightened or released with one hand.

INNOVATION 2

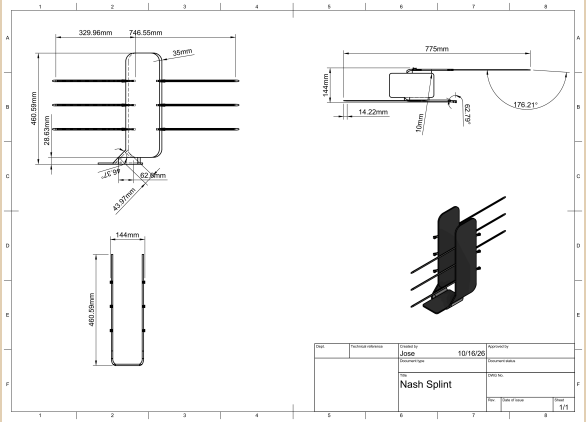
INTEGRATED HYGIENE

A breathable, sanitary design that can be cleaned and checked over weeks of wear.

INNOVATION 3

THERMAL PHASE DESIGN

A TPU element flexible enough to wrap the leg and firm when needed.



Detail drawing, dimensions in mm · Jose, 10/16/2026

YOUR DESIGN · BEYOND THE REQUIREMENTS

MORE THAN THE CHALLENGE ASKED

1. **Three lunar innovations**
   The challenge asks for at least one.
2. **Custom fit from a scan**
   Printed for one patient instead of fixed sizes.
3. **One-handed use**
   The injured astronaut can tighten the straps alone.
4. **Backed by user research**
   A Needs Assessment Survey of 56 students and an interview with our school's athletic trainer.
5. **Four materials tested**
   PLA, ABS, TPU and PETG, with CF-PEEK recommended for flight.

NEEDS ASSESSMENT SURVEY · 56 STUDENTS

Ankle injured most	63%
Can't find a good fit	54%
On their feet a lot	96%
Devices help healing	98%

WHY THE MOON NEEDS IT

A return to Earth takes days, and launching splints in every size wastes launch mass.

$1M launch cost per kilogram	**1/6 g** lunar gravity	**−1–2%** bone density per month